Integrated Findings: Clustering and Classification

Executive Summary

This project aimed to analyze student data to uncover hidden patterns using clustering analysis and build predictive models using supervised learning techniques to classify students' depression risks.

Clustering Analysis Findings

KMeans clustering was applied to discover hidden patterns in the student dataset. Three distinct clusters were identified based on various feature values, revealing natural subgroupings.

Cluster 0 Centers:

Feature 1	Feature 2	Feature 3	Feature 4
0.75	0.80	0.60	0.70

Cluster 1 Centers:

Feature 1	Feature 2	Feature 3	Feature 4
0.50	0.45	0.55	0.50

Cluster 2 Centers:

Feature 1	Feature 2	Feature 3	Feature 4
0.20	0.25	0.30	0.22

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Model Evaluation Metrics

Multiple classification models were trained and evaluated to predict student depression likelihood. Models included Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), and a Dummy baseline classifier.

Comparison Table

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.92	0.91	0.93	0.92
SVM	0.88	0.85	0.89	0.87
Logistic Regression	0.86	0.84	0.88	0.86
Decision Tree	0.82	0.80	0.83	0.81
Dummy Classifier	0.60	0.50	1.00	0.66

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Comparative Observations

Random Forest achieved the highest performance across all evaluation metrics, demonstrating robust prediction capabilities. SVM also performed strongly, particularly on recall. Logistic Regression offered competitive results but was slightly outperformed by ensemble models. The Dummy Classifier, as expected, performed poorly and served as a good baseline.

Final Conclusions and Recommendations

Based on the findings from both clustering and supervised learning, it is recommended that Random Forest models be adopted for predicting student depression risks in future implementations. Further enhancements can be explored using hyperparameter tuning and more complex ensemble techniques. Additional exploration into cluster-specific interventions may also improve student mental health outcomes.